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Agency in Human-AI Collaboration for Image Generation and Creative Writing: Preliminary Insights from Think-Aloud Protocols

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ABSTRACT

The integration of generative artificial intelligence into creative domains raises critical questions about human agency in AI-assisted creation. In this two-study article (study one, $n = 6$; study two, $n = 7$), we explored how people experience creative agency during co-creative tasks with AI, focusing on image generation and writing. Using think-aloud protocols and post-task semi-structured interviews, we conducted an analysis to identify four central dimensions of agency: creative self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative product, as well as the self-regulatory and metacognitive mechanisms that support sustaining agency. Preliminary findings reveal that agency in human – AI co-creation fluctuates across the creative process. Participants employed adaptive strategies – such as progressive refinement, selective appropriation, and counter-inspiration – to reinforce agency when navigating unpredictable AI outputs. However, moments of diminished control and uncertainty about ownership were common, particularly when AI contributions deviated from user expectations. To contextualize these experiences, we applied the Co-Creative Framework for Interaction Design, highlighting how tool affordances shaped participants' sense of agency. Participants' shifting experiences of agency suggest a need for adaptive system features that align with evolving user goals.

Introduction

The rise of generative artificial intelligence (GenAI) has fundamentally reshaped how individuals approach the act of creation (Mao et al., 2024; Oppenlaender, 2022; Rafner et al., 2023a; Vinchon et al., 2023; Wadinambiarachchi et al., 2024). Moving beyond structured and repetitive tasks, AI can now play an active role in creative processes, collaborating to produce art, music, literature, and other creative outputs that were once the exclusive domain of human creators (Cano-Marin, 2024; Chiu et al., 2023; Lee, 2022; Mai, 2024; Schuckart, 2023; Takaffoli et al., 2024; Voola et al., 2024). This shift from human-only to hybrid creative processes has made human-AI co-creativity – a collaborative process between human creators and AI systems – a focal point of scientific exploration (Davis & Rafner, 2025; Davis et al., 2025; Kantosalo & Takala, 2020; Karimi et al., 2018; Rafner, 2025; Rafner et al., 2020; Rezwana & Maher, 2022). This evolving relationship raises critical questions about the

involvement and perception of the agentic human self in creative processes (Brainard, 2023) but also about how AI systems can be designed to enhance, rather than diminish, human agency.

Agency is characterized as the intentional and value-driven enactment of creative processes through self-reflective critical judgment (Doshi & Hauser, 2024; McGuire et al., 2024). Drawing on this agentic perspective rooted in creativity research (Karwowski & Beghetto, 2019; Karwowski et al., 2019), we investigated what role individual agency plays over the course of the co-creative process. Agency in human-AI collaboration likely fluctuates across different stages of the creative process, shaped by both the design of AI systems and the evolving needs of users (Watkins & Barak-Medina, 2024). In this paper, we present analysis of think-aloud protocols and post-task semi-structured interviews for expressions of agency as humans interacted with AI tools in two co-creative tasks. By means of a thematic analysis, we identified four central dimensions of agency

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that varied across the creative process: creative self-efficacy, control over creative action, autonomy in the creative process, ownership of the creative product, and the self-regulatory and metacognitive mechanisms that support sustaining agency. The indicators of agency formed the basis for coding the think-aloud protocols. Importantly, the findings could provide broader implications for Human-Computer and Human-AI Interaction, offering preliminary suggestions how AI tools could support dynamic shifts in agency (Davis & Rafner, 2025; Davis et al., 2025; Mao et al., 2023; Weisz et al., 2023).

Alongside qualitative coding of these think-aloud transcripts, we apply the Co-Creative Framework for Interaction Design (CoFI; Rezwana & Maher, 2022) in the interpretation to take into account each tool's affordances and interaction styles. By examining differences across interaction modes, we seek to explore how specific features of AI tools may relate to perceptions of agency. This dual approach provides preliminary insights on the evolving interplay between system design and the user's creative self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative product. Rather than assuming human-AI creative agency to be fixed or uniform, our research probes the possibility that it may be dynamic and context-dependent, shifting in response to both self-regulation and metacognitive strategies and tool affordances.

Creativity as agentic action in human-AI co-creation

Creativity is experiencing a significant digital transformation (Ceh et al., 2024; Rafner, Beaty, et al., 2023), evidenced by the rise of GenAI models that often match or even exceed human performance in certain creative tasks (e.g., Haase & Hanel, 2023; Hubert et al., 2024). Humans still have the upper hand in more complex creative tasks, (e.g., Chakrabarty, Laban, et al., 2024; Rafner, Beaty, et al., 2023) and scholars emphasize that human agency remains central to the creative process, particularly in human-AI co-creativity (Brainard, 2023; Guo et al., 2023; Moruzzi & Margarido, 2024; Watkins & Barak-Medina, 2024). AI is not a replacement for human creativity, but a resource that can either amplify or constrain an individual's self-efficacy (McGuire et al., 2024; Newman et al., 2024), autonomy in the creative process (Benford et al., 2024), control over creative action (Benford et al., 2024; Chakrabarty, Laban, et al., 2024), and ownership over the creative product (Lovato et al., 2024) depending on how the interaction is structured (Davis et al., 2017, 2025; Rezwana & Maher, 2022).

Creativity emerges from the dynamic interaction of ability (cognitive potential), personality, such as openness and self-beliefs, and domain-specific expertise (Benedek, 2024) within a process shaped by environmental influences, ultimately leading to the production of novel and useful outcomes as recognized within a social context (Plucker et al., 2004). While everyone possesses some level of potential to think and behave creatively (e.g., Kaufman & Beghetto, 2009; Runco, 2004), not all individuals pursue or persist with creative efforts (Kuratomi et al., 2023; Lucas & Nordgren, 2015). Researchers have increasingly recognized that realizing creative potential hinges on an individual's choice to act, framing creativity as an agentic behavior rooted in personal beliefs about competencies (Karwowski & Beghetto, 2019; Sternberg, 2003). From this perspective, creativity is not merely an innate trait but a skill to be intentionally cultivated through effort and self-direction. This agentic view raises a key question: How does the sense of agency fluctuate during the creative process? The rise of GenAI prompts concerns about user control, autonomy, shifts in ownership, and whether AI strengthens or undermines creative self-confidence (McGuire et al., 2024; Tankelevitch et al., 2024). As AI becomes a prominent creative partner, scholars argue that traditional notions of agency must be reevaluated to account for its evolving role in human-AI co-creation (Moruzzi, 2022).

The perception of AI as either a supportive or competitive force depends on how its contributions interact with human input (Mao et al., 2023, 2024). While AI can enhance agency by generating ideas and expanding creative potential, it may also diminish users' sense of ownership and control if its influence overshadows their contributions (Chakrabarty, Padmakumar, et al., 2024). A primary factor in fostering agency is creative confidence – or self-efficacy – which is central to feeling capable within the creative process (Bandura, 1997, 2001; Karwowski & Beghetto, 2019; Zielińska et al., 2021). According to Bandura's social cognitive theory, individuals must believe in their ability to succeed in order to take action (Bandura, 1997, 2001; Karwowski et al., 2019). Research shows that this self-belief is essential in bridging creative potential and behavior (Chen, 2016; Lebuda et al., 2021). When individuals feel agentic, they take initiative, make decisions, intentionally choose the direction, and exert control over their creative work (Bandura, 2001, 2006; Code, 2020). In human-AI co-creativity, creative self-efficacy plays a crucial role in shaping creative performance (McGuire et al., 2024). Newman et al. (2024) argue that GenAI can support the development of creative self-efficacy, particularly when users engage in iterative adjustments.

Two fundamental dimensions of agency essential for motivating creative action are control and autonomy (Klein, 1989). Control refers to the ability to direct actions and make decisions, while autonomy involves the freedom to choose one's creative path and express unique ideas. In traditional individual creative tasks – using tools like pen and paper or text editors – both control and autonomy are often assumed. Creators maintain direct oversight of their process, freely navigating their creative decisions. However, GenAI systems introduce novel interaction dynamics, such as turn-taking, initiative timing, and responsiveness, which can influence the extent to which users feel in command of their creative process (Davis et al., 2025; Rezwana & Maher, 2022). This shift raises critical questions: How much control and autonomy do humans retain when co-creating with AI? How do they navigate moments of diminished agency? And what strategies help them regain a sense of control (Chakrabarty, Laban, et al., 2024; Mulatti & Treccani, 2023)?

Although creative self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative product are conceptually distinct constructs, they are closely interrelated. For example, autonomy and self-efficacy affect creative performance (Chiang et al., 2022). In educational contexts, autonomous learning fosters creative cognition primarily through increased self-efficacy as a mediator (Orakci, 2025). Similarly, autonomy support has been shown to strengthen self-efficacy beliefs (Ito et al., 2024). Experienced control has been identified as a primary antecedent of psychological ownership (Gardner et al., 2023; Pierce et al., 2003). Moreover, psychological ownership positively correlates with self-efficacy (Avey et al., 2009). A significant relationship has also been observed between cognitive control and psychological ownership, which mediates effects on engagement in organizational settings (Mayhew et al., 2007). Although there is a lack of comprehensive models linking all these dimensions together, it can be assumed that they matter and interact throughout the creative process. Testing the relation between these four agency-related constructs could be a crucial next step toward better understanding how people experience agency in co-creative processes with AI.

The design and responsiveness of AI systems significantly shape people's creative experiences. For instance, when AI tools act solely upon explicit user prompts, users maintain a stronger sense of control. Yet, when AI autonomously generates or unpredictably alters content, users may feel their control slipping, perceiving the AI as an intrusive collaborator rather than an assistive tool (Bishop & Erden, 2012; Chakrabarty, Laban, et al.,

2024). This sense of diminished agency can lead to frustration, challenging traditional expectations of ownership and authorship in creative work. Consequently, understanding how to design AI systems that respect and enhance human agency is crucial for fostering productive and satisfying human-AI creative collaborations.

Closely linked to the concepts of control over creative action and autonomy in the creative process is ownership of the creative product – a sense of authorship and the feeling that an idea or product belongs to the person who created it (cf. Pierce et al., 2001). While most contemporary creativity literature does not explicitly address self-perceptions of ownership when incorporating external sources of information, the topic has garnered attention with the widespread adoption of genai (Chakrabarty, Padmakumar, et al., 2024; Lovato et al., 2024; Reza et al., 2025; Weber et al., 2024) and is indeed important for creative behavior (Zhang et al., 2024). Discussions on ownership of creative outcomes are not entirely new; Historical examples highlight similar tensions: Swedish painter Hilma af Klint attributed her work to spiritual guidance, prompting debates on authorship (Burton, 2021), and artist colonies saw master artists take credit for co-created works (Efland, 1990). This parallel resonates with contemporary users' struggles to reconcile their creative contributions with those of the AI (Eshraghian, 2020; He et al., 2025; Kantosalo & Takala, 2020; Li et al., 2024), questioning where the line between “mine” and “ours” should be drawn. Design choices significantly impact ownership. Interactive, back-and-forth ideation (“ping-pong”) can foster greater creative control than fully automated AI generation, leading to a higher sense of ownership (Reza et al., 2025). Aligning with Human-Centered AI (HCAI) principles, balancing automation with user contribution oversight is key to preserving ownership in human-AI co-creation (Mao et al., 2023; Sherson et al., 2024; Shneiderman, 2020; Weisz et al., 2023).

While creative self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative product provides a necessary foundation for making decisions about starting, continuing or ending creative action, effective agency within the creative process also relies on mechanisms that help individuals regulate and monitor creative activity: generating ideas and executing the plan (Lebuda et al., 2024). These mechanisms, creative metacognition and creative self-regulation, further support agency by enabling creators to direct their cognitive, emotional and social resources (Zielińska et al., 2021, 2024). Creative metacognition involves knowledge related to creative cognition, monitoring and controlling one's cognitive processes related to creativity (Lebuda & Benedek, 2023). This includes

evaluating ideas and performance, selecting effective strategies for idea generation, and determining when to conclude a task (e.g., Benedek & Lebudá, 2025; Urban & Urban, 2023). Recent studies on human-AI co-creation and metacognition have pointed out that while cognitive demands decrease as users offload tasks, meta-cognitive demands increase (Risko & Kelly, 2023; Tankelevitch et al., 2024). While metacognition focuses on managing cognitive processes, creative self-regulation involves planning actions, overcoming obstacles, managing emotions, and fostering a willingness to share creative output (Ivcevic & Nusbaum, 2017; Ivcevic et al., 2023; Zielińska et al., 2021). When self-efficacy, control, autonomy, and ownership drive motivation for engaging in creative actions, metacognition and self-regulation determine how individuals execute these actions (Karwowski et al., 2019). An important mechanism of self-directed, goal-oriented action that supports effective creative performance involves the use of meta-cognitive and self-regulatory strategies. These strategies are beneficial when solving creative tasks – whether through mentally manipulating objects or ideas, taking different perspectives, searching memory, or drawing on past experiences (Gilhooly et al., 2007; Jankowska et al., 2018; Rubenstein et al., 2020). However, it remains unclear whether people employ different mechanisms when collaborating with AI that help them effectively manage the process and sustain their sense of agency.

To systematically analyze how different tool designs influence these dimensions of agency, researchers have developed frameworks that map the interaction dynamics of co-creative systems. One such model is the Co-Creative Framework for Interaction Design (CoFI; Rezwana & Maher, 2022), which provides a dual-axis lens for understanding human-AI creative collaboration. CoFI distinguishes between two levels of interaction: collaboration between the human and AI, encompassing turn-taking, communication style, and the timing of initiative; and interaction with the shared creative product, such as opportunities for iterative refinement, direct manipulation, or generative extension. This framework helps illuminate how different affordances embedded in AI tools can shape users' autonomy, control, and ownership – thereby offering a structured way to interpret agency in human-AI creative processes.

The present study

The present study examines how agency is experienced in human-AI co-creative tasks by focusing on four central pillars which were determined via an abductive coding process (see methods for details) – creative self-efficacy, control over creative action, autonomy in the

creative process, and ownership of the creative product. Each pillar offers a perspective on how people navigate and regulate the creative process to sustain agency within human-AI collaborative creative contexts. Specifically, we address the following research questions:

RQ1: How are agentic creative self-beliefs expressed over the creative process during collaboration with AI?

RQ2: How do people regulate their creative agency – through self-regulation and metacognitive strategies – when collaborating with AI?

The research is divided into two parts, each examining a different modality of human-AI co-creation: Study 1 explores human-AI co-creative image generation, while Study 2 investigates human-AI co-creative writing (text generation). Writing and image generation were chosen as focal points because they represent distinct yet complementary modes of interaction in generative AI, both widely adopted in creative fields (Anantrasirichai & Bull, 2022). Each study, detailed below, leverages these interactional levels (between collaborators and with a shared product; Rezwana & Maher, 2022) to investigate how agency manifests within these popular AI-assisted creative tasks.

To investigate how interaction styles influence agency in co-creative tasks, we selected tools that reflect distinct modalities – slider-based image blending, text-to-image generation, and text continuation. These tools were chosen for their differing affordances and because they allow comparative analysis across multiple forms of human-AI communication. For the text generation task, we selected Sudowrite (detailed below) over more general-purpose models like ChatGPT because it is specifically designed for creative writers, offering features such as suggestion curation. This made it particularly well-suited for studying nuanced writer-AI interactions. Using the CoFI framework, we systematically analyzed each tool's interaction characteristics to assess how collaboration style, task distribution, and communication modality shaped participant experiences of self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative product. As shown in Table 1, all tools followed a turn-taking participation style in which users initiated input and the AI responded. While text input and direct manipulation were the primary modes of intentional communication from users to AI, feedback from the AI was indirect – either visual or textual. Notably, Sudowrite's design enabled evaluative and extension-based contributions, making it more collaborative than the purely generative image tools.

Table 1. Summary of CoFI analysis of the AI tools used in the studies.

Interaction between Collaborators							
Tool	Collaboration style				Communication style		
	Participation style	Task Distribution	Timing of Initiative	Mimicry	Human to AI Intentional Communication	Human to AI Consequential Communication	AI to human Communication
Slider based image generation (crea.visions)	Turn-taking	Same Task	Planned	None	Direct manipulation	None	Indirect visual feedback
text-to-image	Turn-taking	Same Task	Planned	None	Text input	None	Indirect visual feedback
Text generation sudowrite	Turn-taking	Same Task	Planned	Mimic	Text input, Direct manipulation	None	Indirect text feedback
Interaction with the Shared Product							
Tool	Creative Product						
	Contribution Type		Contribution Similarity		Creative Process		
Slider based image generation (crea.visions)	Create new, Refine, Extend, Transform		High (Pleasing)		Generation		
text-to-image	Create new, Extend, Transform, Refine		High + Low		Generation		

(Continued)

Method

Ethical approval was received from Aarhus University prior to the data collection, and informed consent was obtained from all participants.

We conducted two qualitative studies using a mixed-methods approach that combined think-aloud protocols and semi-structured interviews. The sample size of each study aligns with the concept of discount usability (Lewis & Sauro, 2021) which supports using small, focused samples to uncover interaction patterns in early-stage research. Though originally developed to identify interface issues, these methods are also used in exploratory qualitative studies to examine use strategies and lived experiences (Calderwood et al., 2020; Clark et al., 2018). We adopt this approach by combining within-subjects comparisons with rich process data – think-aloud protocols and interviews – to explore how participants navigate agency across co-creative tasks with AI. This methodological stance aligns with human-computer interaction research that uses small-sample, process-focused studies to explore emergent user strategies (Blandford et al., 2016; Lazar et al., 2017; Rogers et al., 2011) and is consistent with qualitative psychology approaches that emphasize depth over breadth in exploratory contexts (Braun & Clarke, 2006; Guest et al., 2012).

Both studies employed the think-aloud method, where participants verbalized their thoughts during task performance, providing insights into their decision-making processes and interactions with AI tools (Fleck & Weisberg, 2004; Pringle & Sowden, 2017). Additionally, semi-structured interviews were

conducted after the tasks to explore participants' perceptions of creative agency, the AI's role in collaboration, and their overall experiences with the tools. Additional survey data were collected on topics such as whether users perceived their creations as art; however, this was not included in our analysis as it fell outside the study's primary focus on agency in human-AI collaboration.

Participants

Table 2 presents relevant information on the participants from Study 1 and Study 2.

For study 1, participants were recruited through convenience sampling. Participants were from diverse backgrounds, including students, professionals, and the general public. Additionally, they were required to be fluent English speakers and to have a personal computer they would bring to the experiment. For study 2, participants were recruited through contacting local creative writing groups with information about how to sign up for the study. Participants were required to have some experience writing fiction in English, and to have normal or corrected vision.

Study 1: image co-creation

Tools used

In this study, we used two different AI enhanced image generation tools:

Table 2. Sample size, age and gender distributions, educational backgrounds, experiences related to creativity and AI tools, as well as language proficiency of participants in study 1 (image co-creation) and study 2 (text co-creation).

Characteristic	Study 1 Participants	Study 2 Participants
Sample Size (N)	6	7
Age (Mean $\pm$ SD)	27.3 $\pm$ 5.4 years	27.0 $\pm$ 4.9 years
Gender Distribution	4 female, 2 male	3 female, 3 male, 1 other
Educational Background	All with higher education	All with higher education
Creative Experience	4 engage in weekly creative activities (e.g., photography, drawing)	Creative writing experience ($M = 10.1$ years; $SD = 7.3$)
Familiarity with AI Tools	All heard of image generation, 3 had experience with it	All familiar with LLM chatbots but no specific writing tool experience.
Languages	English spoken fluently by all (1 native speaker)	English spoken fluently by all (1 native speaker)

Crea.visions is an image *blending* tool (Rafner et al., 2021; Rafner, Zana, et al., 2023) which uses a pre-trained StyleGAN (Karras et al., 2020) to allow users to blend images based on two components: style and content. Style is defined as small-scale features and texture, while content refers to the large-scale features of the image. The users adjust these by using sliders for the style and content of each image. They can blend up to four images together which they choose from a gallery of 102 images. The images contain a variety of both color palettes and motivic elements, such as water, buildings, mountains, etc (see Figure 1a).

Stable Diffusion is a text-to-image¹ latent diffusion model designed by engineers and researchers from CompVis, Stability AI, and LAION (<https://stability.ai/>). It uses an open-source model for learning visual concepts from natural language, which is directly integrated into the AI model and the embeddings are processed by the diffuser models used (Borji, 2022). It uses an open-source model for learning visual concepts from natural language, which is directly integrated into the AI model and the embeddings are processed by the diffuser models used (Borji, 2022). For the purpose of this experiment, an open-source Hugging Face script was integrated into a web interface (see Figure 1b).

Figure 1a shows the image blending interface. The users blend the four images in the bottom by adjusting the sliders underneath them. The generated image is shown in larger size on the top. Participants were instructed to ignore the additional UI elements. Figure 1b illustrates the text-to-image interface, where on the top of the page, the instructions are displayed for the participants to write their prompts in the text field, and then use the “generate image” button for creating it. The participants received two tips to help the prompt writing process: Tip 1: *feel free to be concrete and descriptive and explore anything that comes to your mind.* Tip 2:

you can specify the style of the created image (e.g., impressionist, vaporware, watercolor, salvador dali, anime, movie screenshot, etc.).

Procedure

Participants were introduced to the study before filling out the consent form and pre-survey using SurveyXact.² Then, three of the participants (Group 1, consisting of P02, P04, P05) began the slider-based image blending tasks, while the other 3 (Group 2, consisting of P01, P03, P06) began with the text-to-image tasks. Once the participant opened their first tool, the researcher introduced the study in more detail and instructed the participant to be creative, to think aloud, and informed them about the time limit (10 minutes per task) and given their first task. They also received a brief oral introduction to each tool. Participants were given three tasks with both image generating tools. After using the tools, participants took part in semi-structured interviews designed to explore their experiences. Figure 2 shows an overview of the study procedure.

Tasks & instructions

When choosing tasks for participants to create images, three factors were considered: limited content in the image blending tool, time constraints, and the need for variety. The final tasks were: 1) **Create a comforting image**, 2) **Illustrate the following quote**: “From that time on, the world was hers for the reading. She would never be lonely again, never miss the lack of intimate friends. Books became her friends and there was one for every mood.” – *A Tree Grows in Brooklyn* by Betty Smith (1947), and 3) **Design an indie pop album cover** while listening to an indie pop song as background music *Orpheus* by The Beaches; *Hey Doode* by Peach Fuzz. The first task was the most abstract, while the second provided more constraints. The third task (album cover) was accompanied by music from the selected genre, as some participants were unsure what indie pop meant.

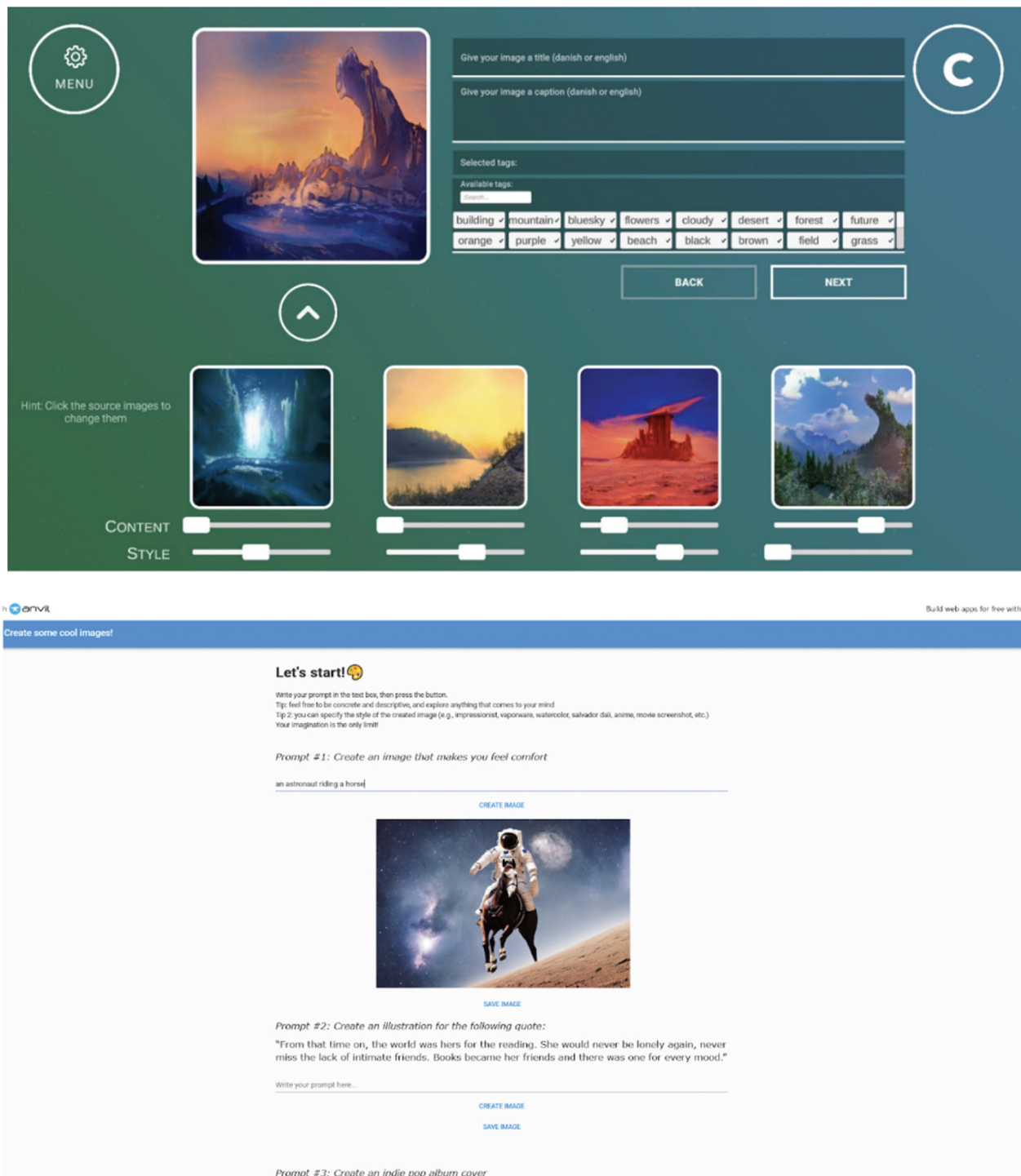


Figure 1. Screenshots of the image blending (a) and text-to-image (b) interfaces.

Study 2: text co-creation

Tools used

We used the “Write” mode in Sudowrite, a popular subscription-based AI tool for creative writers (Sudowrite, n.d.). This is a form of continued auto-writing where the text preceding the cursor placement is automatically used as input to the AI. After an AI query (a “Write” button

press), the underlying LLM generates suggestions on how to continue the text. To tailor the AI output, certain parameters can be changed, such as the number and length of AI suggestions. Additionally, a “creativity” setting is linked to the temperature parameter of the LLM and controls how unexpected the AI generations can be. The settings for these parameters were decided based on

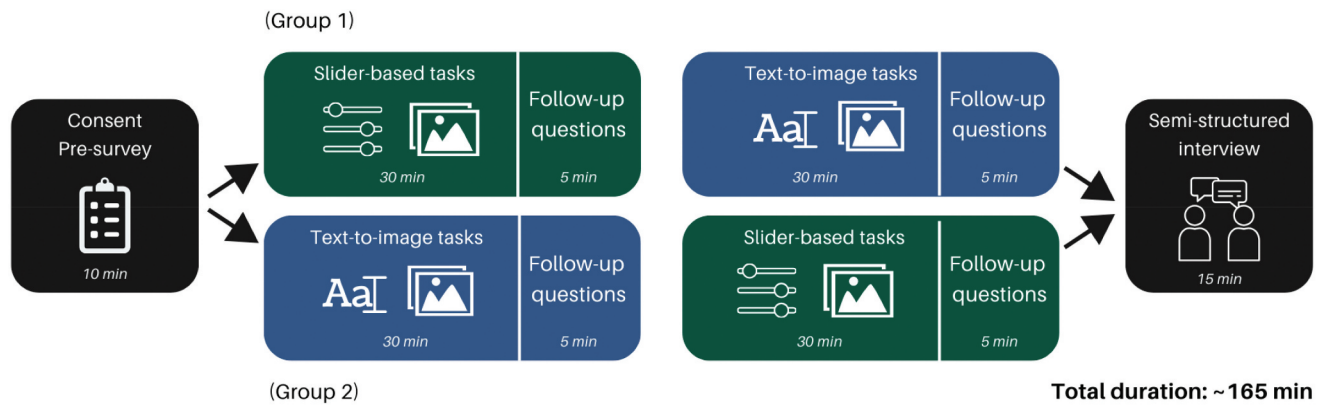


Figure 2. Illustration of the experimental procedure of study 1.

previous literature (Calderwood et al., 2020; Singh et al., 2022) and a pre-pilot study: 2 AI suggestions, ~50 word AI output length, and *medium* creativity.

Figure 3 illustrates the Sudowrite interface in the second study: 1. The text editor where the user can input and format text. 2. The “Write” button. When it is clicked, two cards appear on the right with AI text suggestions, continuing from where the cursor was when the button was clicked (in this example, it was in the middle of a sentence). 3. The AI suggestion history where new AI suggestions show up at the top, each on its own card. Previous AI suggestions can be found by scrolling down through the history. The user has the option to copy the AI suggestion, or to directly insert it

into the text editor. The rating buttons (thumbs up/down and star symbols) were not used in the study, and participants were told to ignore them.

Procedure

Figure 4 shows an overview of the study procedure. Participants were introduced to the study before filling out the consent form and pre-survey. Then, four of the participants (Group 1, consisting of P01, P03, P05, P07) began the non-AI writing session, while another three (Group 2, consisting of P02, P04, P06) began the AI writing session.

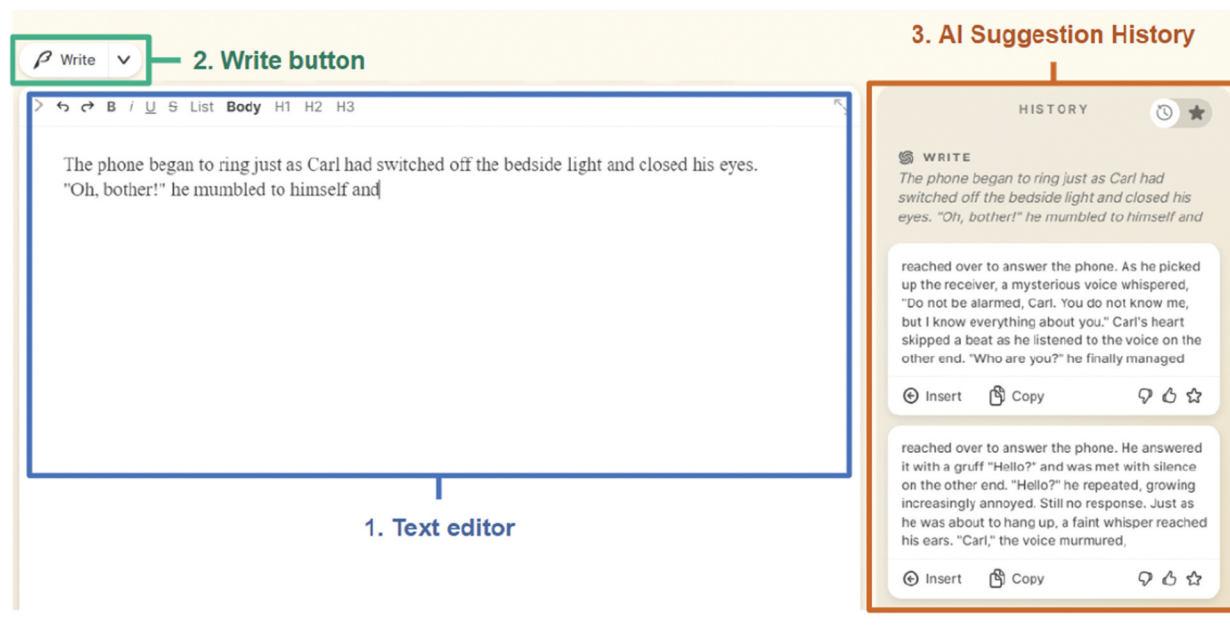


Figure 3. The Sudowrite writing interface for the AI writing session (Hansen & Rafner, 2025).

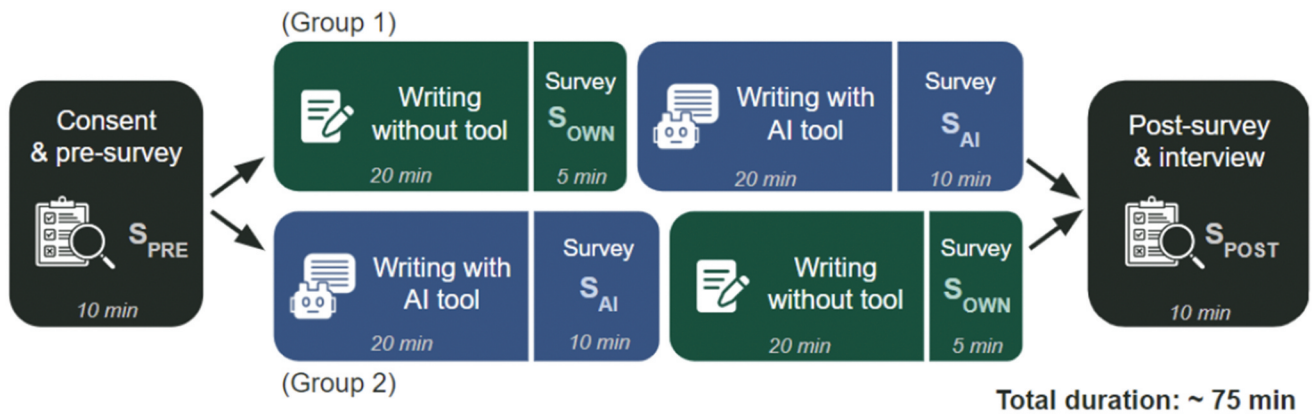


Figure 4. Overview of the experimental within-participants study design in study 2.

Task & instructions

For each writing session, participants were instructed to write a creative short story, and were given a small prompt which they had to continue. The prompts *A train arrives at the station* and *The phone began to ring* were counterbalanced across AI and non-AI sessions. To ensure streamlining of instructions, participants watched a short tutorial video before the AI writing session. The video was created for the study to explain the interface and how to use the AI tool. Participants had 20 minutes to complete each task.

The two writing prompts have been used in previous research (Singh et al., 2022) and were chosen because they are action-oriented, giving participants a starting point for their writing, and they are open-ended, letting participants steer the story in many possible directions.

Analytical approach

To investigate participants' experiences of agency while interacting with the AI systems, we employed thematic analysis, a widely used qualitative method, which allows researchers to identify, analyze, and interpret patterns within data while maintaining

flexibility in theoretical orientation (Braun & Clarke, 2006). We used codebook thematic analysis as it is particularly useful when working in collaborative research teams (Guest et al., 2012) as opposed to single author analysis. Table 3 provides definition examples for each code: creative self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative products that manifested in human-AI co-creative tasks.

The development of these four codes followed a multi-stage, abductive process, with an inductive coding round based on creativity frameworks related to creative metacognition (Lebuda & Benedek, 2023) and creative self-regulation (Zielińska et al., 2022). To apply these codes, the second and third authors independently analyzed all 13 transcripts (six from the image generation study and seven from the text generation study) using Taguette,³ open-source qualitative data analysis tool (Rampin et al., 2021), while the fourth author performed spot-checks for consistency. Regular team discussions allowed the researchers to interpret ambiguous data excerpts and align their coding approach with the agency dimensions under investigation.

All transcripts were analyzed for the occurrence of specific codes, following a structured coding

Table 3. Code, definitions, and examples.

	Definition	Examples
Autonomy in the creative process	Individuals sense a freedom to choose their path and express their ideas.	S1P5 "I'm going to try romanticism instead of Renaissance"
Control over creative action	Involves individuals having a command over one's actions and decisions within the creative process. It's about making deliberate choices and managing the execution of creative ideas.	S1P2 "I feel like I have no agency because it's more like I'm telling this machine and it's just surprising me"
Creative Self-Efficacy	Refers to an individual's belief in their own ability to successfully execute the actions required to achieve specific creative goals. It's about having confidence in one's skills and abilities.	S2P6 "I know how large language models work, I was able to use it in my favor for writing something that is a bit more unexpected"
Ownership over the creative product	Refers to an individual's belief that they have contributed a significant portion of effort to the final product and thus feel like they are responsible for it.	S2P1. "Like, the ideas are mine. I didn't get any new ideas about the content [from the AI]."

approach (e.g., Saldaña, 2015). The identified codes were then transferred to an Excel table, where their occurrence was examined. This approach aligns with previous research on qualitative coding and analysis in cognition studies (e.g., Ericsson & Simon, 1993; Gero & Kannengiesser, 2012). By mapping the codes in relation to the creative process timeline, we aimed to capture patterns of creative agency fluctuation (e.g., Sawyer, 2012; Wallas, 1926).

Materials

The semi-structured interviews, additional coded excerpts, and information on the coding procedure can be found here: https://osf.io/u7nya/?view_only=2c323ed6d7384669a17b6952aedf6d42

Results

Expressing creative agency in human – AI collaboration

We now present three vignettes from think-aloud protocols and post-task reflections, illustrating how participants navigated creative self-confidence, control over creative action, autonomy in the creative process, and ownership of the creative product within human-AI co-creation. These examples are analyzed in order to address how participants express agentic creative self-beliefs throughout the creative process (RQ1), and how they employ self-regulation and metacognitive strategies to regulate their creative agency (RQ2).

In Vignette 1, as seen in Figure 5, a participant using the slider-based image tool initially exercised autonomy

in envisioning a cozy fireplace scene. However, they soon experienced limited control and disrupted ownership, as the tool's constraints prevented fine-tuning the image to match their vision. The AI's output, while pleasing, failed to accommodate their deeper preferences, reflecting a lack of consequential communication and stylistic mimicry. Frustrated but determined, the participant pivoted creatively – demonstrating creative self-efficacy through their willingness to adapt, and more broadly drawing on their agentic creative self-beliefs to reassert direction and intention in the co-creative process (RQ1).

As seen in Figure 6, while writing a story using the text generation tool, participant 6 received AI suggestions that leaned on a familiar trope – train delays evoking anxiety. Rather than following this predictable path, the participant felt autonomy and exercised strategic control by intentionally steering the narrative in the opposite direction. Viewing the AI's output as counter-inspiration, they used it as a springboard to craft a relaxed, unexpected scene.

This oppositional strategy reflected high creative self-efficacy, supported by the participant's familiarity with language models and their tendency toward common patterns. By treating the AI's suggestions as flexible, even provocative inputs, the participant maintained authorship and shaped the story on their own terms. This response drew upon broader agentic creative self-beliefs – including a sense of autonomy and intentionality – to diverge from the AI's expected outputs. This vignette illustrates how AI-generated content can prompt creative divergence, not only through acceptance but also through deliberate resistance (RQ1).

Vignette 1 (S1P3, 1:05:21) - slider based image generation

Participant information: *Expert in AI; no drawing experience and does not engage in creative activities; strong English proficiency and weekly use of AI language models*

Context: After the task was completed, the participant is reflecting on how they created an image with the slider-based tool for the prompt: *create an image that makes me feel comfortable*

Participant think-aloud:

“So, I was trying to think about what places in nature I would feel comfortable, what setting in nature would make me feel comfortable. My original thought was a fireplace camp feeling. I did see an original image that had this, you were in darkness, sitting around the fireplace [+AUTONOMY]. But I didn't feel like I could really add anything to it that made it more like that [-CONTROL]. So, it didn't feel like I could make it mine [-OWNERSHIP]. So, I tried to go in a different direction [+EFFICACY].”

Final image



Figure 5. Vignette 1: complex co-existence of autonomy, control, ownership, and efficacy. After the section where the code was present, the code will be written in brackets. A plus sign (+) will denote the presence of the coded phenomenon, e.g., the participant is feeling autonomous, and a minus sign (-) will denote instances where the participant is expressing the lack of said code, e.g., feels lack of control over the process.

Vignette 2 (S2P6, 0:05:36) - writing
Participant information: Experienced fiction writer (7 years) with strong English proficiency and weekly use of AI language models

Context: During the task continuing to the prompt “A train arrives at the station...”, the participant pressed the button to get AI suggestions for the first time.

Participant text	AI suggestions	Think aloud
Before AI suggestions: <i>A train arrives at the station where it was not supposed to arrive. He was expecting that. Problems always happen when you travel using the almighty German Deutch Bahnhof[sic!].</i>	1: “As the train screeched to a stop, Marius couldn’t help but feel a sense of dread wash over him. He checked his watch and cursed under his breath. He was already running late for his meeting and this was not helping his cause. Sighing, he grabbed his briefcase and stepped out onto the platform” 2: “As John stepped off the train, he couldn’t help but feel a sense of unease. He had been on edge since he boarded the train in Berlin, and the unexpected stop only heightened his anxiety. Looking around, he realized	“It's not how I would continue the story. No. [+AUTONOMY] Because I think since it is about train arriving times and stations, it is creating the feeling of anxiety or yeah, “checked his watch and cursed under his breath.” So it's pretty much based on running late, but I would not continue the story by bringing up this anxiety, this Oh I’m running late, but more like, oh, this was expected. [+CONTROL] I never expect to arrive at my destination when using this train service. So actually, it's giving me the idea of doing the opposite, of what the language model is probably expecting. [+EFFICACY] It's based on people's association with running late and trains and so on, and giving this anxiety. So giving me the idea of doing the opposite. So the story is a bit more unexpected.”
After AI suggestions: <i>[...] However, this time was different. People were not feeling anxious. People were just relaxed, some lost in the realm of music, some lost in never-ending conversations. People were enjoying the journey, and the destination did not matter.</i>		

Figure 6. Vignette 2: autonomy and control over AI input.

Vignette 3 (SIP5, 0:38:16) - text-to-image generation

Participant Information: 5+ years of experience in drawing (digital and traditional), engages regularly in creative activities (most days of the week) with strong English proficiency. Education in design, limited experience with AI image generation

Context: Participant is working with task 2 using text-to-image generation, illustrating the quote: "From that time on, the world was hers for the reading. She would never be lonely again, never miss the lack of intimate friends. Books became her friends and there was one for every mood." — *A Tree Grows in Brooklyn* by Betty Smith (1947). They have an image in mind they wish to create, and they try to achieve it using the text-to-image tool.

Text prompt

A girl laying on a pile of books laying on her stomach, resting her head on her arm renaissance art style

AI generated image**Think aloud**

Okay, see this is what's pissing me off. It specifically says, 'a pile of books,' and this is just one book [-CONTROL]. But other than that, it's actually pretty good.

A girl laying on a pile of books laying on her stomach, resting her head on her arm renaissance art style



Maybe I should write a mountain of books and then if it's going to put her on a mountain, I'm going to lose my shit [-CONTROL].
[waits for image]

A girl laying on a mountain of books laying on her stomach, resting her head on her arm renaissance art style



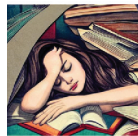
Okay, this is closer to what I was thinking but it looks terrifying. So, I'm going to try with a mountain of books. [+EFFICACY]
I wrote mountain instead of pile. [waits for image]

A girl laying on a mountain of books laying on her stomach, resting her head on her arm romanticism art style



Oh, we're getting there.[+AUTONOMY] Okay, I'm going to try this now. I think it's the last adjustment. I'm going to try romanticism instead of Renaissance [+AUTONOMY], because I feel like with Renaissance it's really going towards the -- this is an art style that I feel like they had-- its church related. You know, like the what do you call it icon--And that's really not the part of Renaissance I was looking for. But I don't know how to point it out better.[-EFFICACY] Maybe romanticism is a better choice. Give me something good. [waits for image]
That is not romanticism at all. But subjectively speaking, it's better than the rest of them but -- okay, last time. [waits for image]

A girl laying on a mountain of books laying on her stomach, resting her head on her arm romanticism art style



Okay, well it's certainly not getting better. I don't like that it's a close up. So, I should just mention that it should be a long shot? Like long distance shot like an- angle.

A girl laying on a mountain of books laying on her stomach, resting her head on her arm romanticism art style, far away perspective



It's not exactly-- far away could be. Last one. But this one again this one so far is like the one that. we would say the best. [waits for image]

It's definitely one that I would connect to the prompt -it's not at all what I was thinking about. But looking at it, I think it's perfect for it.

Figure 7. Vignette 3: autonomy amid control challenges.

As seen in Figure 7, a participant illustrating a book quote with a text-to-image tool maintained autonomy in setting their vision but faced control issues when fine-tuning details (e.g., “pile of books” vs. “mountain of books”). Despite persistent refinements, the participant settled on an image that was acceptable but misaligned with their initial vision. This vignette highlights how tool limitations can strain user control, forcing creators to negotiate between personal vision and AI-driven outcomes (RQ1).

Dynamic agency: fluctuation over the creative process

We examine how participants experienced varying degrees of creative self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative product across different phases of human-AI interaction during image and text generation. Our findings reveal that these agency dimensions fluctuate based on initial ideation, iterative experimentation, and final product evaluation. A visual illustration of the main observations from Study 1 and Study 2 can be found in Figure 8.

During the initial phase of the creative process, participants experienced high degrees of autonomy. They described feeling liberated to explore ideas and set a broad creative goal, when envisioning unique images or narrative elements and they felt autonomous as they

navigated wide-ranging aesthetic choices. For instance, S1P3 said “I’m just exploring to find what would look good. So first is a random exploration where I’m trying to find things. It’s a very loose process. It’s almost hoping to be inspired by the random generation.” However, at this stage, control remained underdeveloped: as they interacted with the unfamiliar AI interfaces, they were challenged to translate abstract concepts into precise inputs and outputs.

As participants moved into the middle stages of co-creation, the focus shifted to iteratively refining outputs for closer alignment with their intended goals. This phase highlighted a struggle for control, particularly when AI-generated results diverged from participants’ envisioned directions – a frequently recurring event. However, it must be noted that there were some conflicting experiences in our data. For example, one participant felt creative autonomy even without full control as they reflected: “There was the creativity that’s not often prompted . . . I wasn’t executing anything. Really. It was just the big ideas. And that was pretty fun to just stay in that zone” (S1P1).

In cases where participants wanted to refine specific aspects of an image (e.g. “a stack of books” in Vignette 3), limitations in AI control features, such as inability to isolate or lock certain image elements, led to frustration and diminished control over creative action. Ultimately, this resulted in mixed ownership perceptions, as participants simultaneously felt responsible

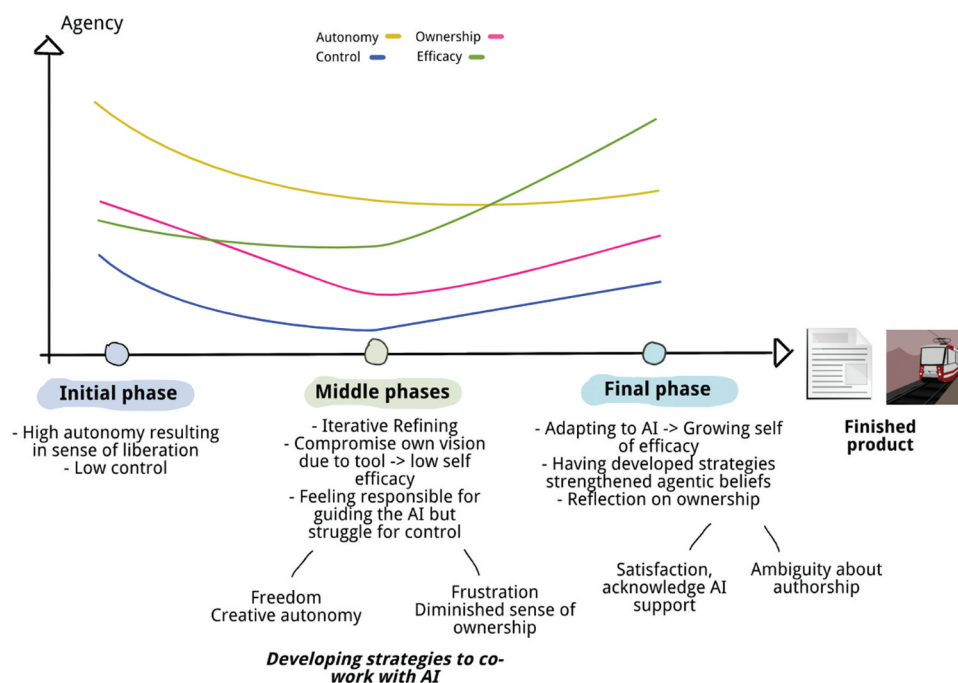


Figure 8. Visual illustration of the main qualitative observations from study 1 and study 2, including dynamic shifts in the four human agency constructs: creative self-efficacy, control over creative action, autonomy in the creative process, and ownership of the creative product.

for guiding the AI yet struggled to claim the final product as fully “theirs.” Advancements in image editing, such as in-painting and out-painting, allow users to add, revise, or extend specific areas of an image while preserving the surrounding content

These disruptions had some participants reflect on ownership long before the final creative product had been created. Specifically, they had to weigh to what extent they could steer the AI tool toward an outcome that would still adequately reflect their own vision while respecting the time constraints of the task and variability of the AI outputs. In these contexts, some participants also indicated struggle by means of low creative self-efficacy, which had them deviate from the initial goal toward adapting what the AI had produced. Although many participants actively set creative goals, not all managed to implement them in ways for them to feel ownership.

In the final stages, participants reported a growing sense of efficacy as they adapted to the AI’s capabilities and limitations. By this stage, many had developed strategies (described below) to either align the AI’s suggestions with their original vision or reinterpret their goals in light of the AI’s output. This adaptive process strengthened their agentic belief in their capacity to manage the creative outcome. Reflective ownership became evident as participants assessed the final product, often with nuanced views about their role. While some felt satisfied and acknowledged the AI’s supportive role, others expressed lingering ambiguity regarding their ownership, questioning how much of the output was truly theirs versus an emergent AI-driven creation.

Is this mAIne?

Although we did not anticipate this in our research question, a close examination of the code occurrences revealed more specific patterns related to ownership. Participants’ reflections post-creation revealed diverse ownership perceptions, shaped by their control, input, and the alignment of AI outputs with their creative intent. Ownership perceptions spanned from viewing

AI as a tool providing raw material to seeing it as a co-creator. This variability was related to participants’ sense of agency and the extent to which they felt in control of the output.

In co-creative writing, several participants described their sense of ownership in spectrum-like terms, suggesting a more fluid perception of ownership during AI-assisted creation. For instance, one participant emphasized their editorial control: “I’m in control of the story. I steered it. I chose what went in and what didn’t” (S2P3). In contrast, others expressed ambivalence, reflecting on the extent to which integrating AI-generated suggestions constituted co-authorship.

Ownership perceptions were related to the degree of congruence between participants’ intentions and AI outputs. Some participants quantified their contribution pragmatically. One noted, “I wrote like 80% of it . . . the paragraph the AI added was generated using my previous words” (S2P6). This aligns with findings suggesting that retaining control over integration preserves a sense of ownership (RQ1).

In AI-assisted image generation, ownership was often linked to the abstractness of the output and participants’ perceived influence over it. Participants felt more ownership when their creative intent was closely reflected. For example, one participant noted, “I feel more ownership over [the abstract ones] because I put more of myself in it by changing the style and making them less realistic” (S1P6). However, some participants found that while abstract images evoked greater ownership, realistic ones seemed less personal, as they resembled preexisting photographs.

Regaining agency: human self-regulation and metacognitive strategies in co-creating with AI

Participants in our study developed several adaptive self-regulation and metacognitive strategies to navigate the complexities of AI collaboration (e.g., unexpected outputs), which can broadly be grouped into three categories, seen in Table 4 and elaborated on below.

Table 4. Self-regulation and metacognitive strategies, definitions, and examples.

Adaptive Strategy	Definition	Example
Progressive refinement	A trial-and-error approach involving incremental iterations to shape outputs over time.	Taking a prompt that already exists and try to amend it
Selective appropriation	Curating and integrating only the AI-generated elements that align with one’s creative goals.	Only taking style elements from an image
Counter-inspiration	Deliberately rejecting or subverting AI suggestions to assert originality or inject contrast.	Doing the exact opposite of what the AI suggests

All these strategies emerged when participants faced challenges during the task.

Progressive refinement: embracing uncertainty and experimentation

Progressive refinement emerged as an adaptive strategy which involved a trial-and-error approach that allowed participants to refine outputs and understand the AI's capabilities incrementally. This was most apparent in the image generation tasks using image blending, where participants manipulated sliders for style and content. While initially challenging – particularly when precise control over elements like a “crying fish” motif (an abstract shape of blob, as interpreted by the participant; to see the image, refer to supplemental material on OSF, Vignette A) was limited – participants persisted in experimenting with settings, refining their work through continuous adjustments. The image blending affordances for Direct Manipulation and Turn-Taking Style feedback were instrumental here, allowing participants to explore modifications visually and incrementally.

This dynamic interaction between participants and the AI tools underscores the role of creative metacognition and self-regulation as outlined earlier. Participants frequently engaged in iterative exploration, as seen when one participant reflected, “I’ll take the prompt that already exists and try to amend it” (S1P3) and later stated, “Let’s see if this works. Hopefully, it’s not any scarier than this” (S1P3). These comments illustrate how participants evaluated and adjusted their strategies to refine outputs, demonstrating the deliberate monitoring and adaptation of cognitive strategies characteristic of creative metacognition.

Selective appropriation: curating AI suggestions to align with personal intentions

Another adaptive strategy, selective appropriation, allowed participants to curate AI contributions, integrating only elements that aligned with their creative vision. For example, one participant noted, “I only want the stars but not the other stuff, so only take the styling” (S1P4), demonstrating a conscious effort to retain specific elements while discarding others. Similarly, another participant, reflecting on the AI's outputs, remarked, “I think the one I had two images ago was probably the one I liked the best” (S1P3), highlighting their ability to balance emotional responses and refine their creative goals, a key aspect of creative self-regulation.

This strategy was particularly effective in text generation tasks with Sudowrite, where participants could assess AI-generated phrases and choose to incorporate only those that served their narrative goals. For

example, one participant (S2P4) appreciated certain stylistic touches provided by the AI, while rejecting suggestions perceived as generic or inconsistent “*I took the verbs and little phrases ... but I think parts of it could be very helpful*”, showing how participants filtered and adapted AI outputs to align with their intentions. Sudowrite's suggestion-based structure facilitated Direct Text Feedback, allowing participants to integrate specific phrases while ignoring others, thus fostering a high degree of control and ownership over the final product.

Counter-inspiration: using AI's predictability to enhance originality

A subset of participants reported deliberately diverging from AI suggestions as a way to enhance originality – what we term “counter-inspiration.” For example, one participant demonstrated this by using an AI-generated text with urgency and tension as inspiration to craft a calmer counter-narrative, stating, “The AI gave me a starting point, but I wanted something completely different” (S1P2). Rather than following AI prompts passively, they viewed AI outputs as reference points to inspire contrasting or unconventional ideas. This strategy was especially evident in Sudowrite's text generation tasks, where the tool's turn-based suggestions allowed participants to select which aspects of the responses to integrate. For instance, one participant used AI-generated text that introduced urgency and tension in a storyline to inspire a counter-narrative of calm. P06 explained, “*The AI expects trains to induce anxiety, so I'll do the opposite*”. Here the participants exercised a high degree of control by taking control over when and how to engage with AI suggestions.

Discussion

Our preliminary findings yielded three key insights: one concerning the dynamic nature of agency, another regarding the strategies participants employed to navigate human-AI collaboration, and a third centered on ownership. While we did not initially anticipate ownership to play such a prominent role, it emerged as a particularly salient theme. In the following sections, we interpret this outcome through the lens of both underlying psychological mechanisms and the affordances provided by the AI tools.

Dynamic agency and tool challenges and affordances

Our first research question asked how are agentic creative self-beliefs expressed over the creative process

during collaboration with AI? We found that participants' agentic self-beliefs could be described through four different constructs: creative self-efficacy, control over creative action, autonomy in the creative process, ownership of the creative product. These facets fluctuated throughout the co-creation process, shaped by challenges posed by the AI systems, tool affordances, and evolving creative goals.

These dynamics reveal that creative agency in human-AI collaboration is neither static nor monolithic. Rather, it evolves in response to the user's stage in the process (e.g., ideation vs. refinement), their confidence in steering the AI, and the system's capacity to accommodate those shifts. Many of the changes participants made during the creative process arose from attempts to navigate or overcome the AI partner's limitations and to regain a sense of agency – an issue explored in more detail in the section *Self-regulation and metacognitive strategies*. Moments when participants could not effectively co-create with the AI, such as being unable to isolate or adjust specific elements, disrupted the flow of creative action. These disruptions often led to frustration, temporary loss of control, and noticeable fluctuations in the trajectory of the creative process. The ability to engage with and adapt to these challenges – rather than being derailed by them – reflects a crucial element of creative self-regulation (Zielińska et al., 2022). Participants' attempts to reorient and restore agency during these moments of breakdown highlight how self-regulatory mechanisms dynamically modulate the creative process in response to external constraints.

The CoFI framework offers a helpful lens to understand how different affordances mediate creative agency. For example, in the image blending task, the immediate feedback through visual adjustments allowed users to exercise Human-to-AI Communication with each adjustment. This iterative exploration enabled participants to adapt their creative expectations, working creatively within the constraints of the tool. Through sustained experimentation, participants often arrived at satisfying outcomes despite technical limitations, reinforcing their efficacy and sense of control over the process. Likewise, Sudowrite's recommendation cards supported Human-to-AI Intentional Communication and participant-driven Timing of Initiative. The recommendation cards enabled participants to decide when to use the AI's predictable output as a prompt for creative divergence, ultimately enhancing their autonomy via applying a conscious strategy (see below for more details on adaptive strategies).

Participants' struggles and adaptations resonate with broader HCI research. Adaptability and enhanced controllability, along with model customization, and

prompt engineering tools, are key features of human-centered co-creative tools (Ko et al., 2023). Although powerful new image-editing techniques – such as inpainting, out-painting, and semantic control (Avrahami et al., 2022; Nichol et al., 2021; Saharia et al., 2022; Yang & Ruhaiyem, 2024) – enable more granular control, they were excluded from our experimental tool to avoid overwhelming users with complex features (Darejeh et al., 2024).

Although our study is preliminary, it seems that participants' shifting experiences of agency suggest a need for adaptive system features that align with evolving user goals. Genai, when properly scaffolded, can support both divergent and convergent thinking in creative production and problem solving (Hoßbach & Isaksen, 2025; Rafner et al., 2020). For example, depending on whether a user is exploring ideas or finalizing a narrative thread, AI systems might alternate between offering more *divergent* or *convergent* suggestions. This distinction mirrors the “temperature” parameter in language models: lower values promote more predictable outputs (convergent), while higher values generate greater diversity (divergent) (Shih et al., 2023). User controls (e.g., global/local toggles for types of help, initiative sliders, “lock style” switches) are essential companions to adaptive behaviors, because they let users shape when divergence/convergence is even allowed to enter the loop (Reza et al., 2025).

Commercial tools like Sudowrite and Jasper (n.d.)⁴ have started to address this by offering brainstorming and refinement modes. However, these tools rely on manual switching. Future systems could integrate reinforcement learning to personalize AI behavior dynamically, adjusting suggestion style, initiative level, or mode based on interaction patterns. This direction aligns with HCAI principles emphasizing user control, transparency, and adaptive support (Shneiderman, 2020) and recommendations for persistent, user-auditable functions that preserve authorial control while enabling selective delegation (Reza et al., 2025).

Self-regulation and metacognitive strategies

Our second question asked how do people regulate their creative agency – through self-regulation and metacognitive strategies – when collaborating with AI? Our answer highlights that the regulation of creative processes in human-AI collaboration is dynamic and multifaceted, characterized by fluctuating dimensions of agency, often shaped by the affordances of the tools and the degree of control they had over specific

functionalities. We identified three strategies specific to cooperation with AI, through which participants actively shaped their creative output, balancing human intention with AI influence: counter-inspiration, iterative exploration, and selective appropriation.

The three strategies identified in our study can be understood as context-specific expressions of broader metacognitive and self-regulatory mechanisms described in previous creativity research (Gilhooly et al., 2007; Jankowska et al., 2018; Rubenstein et al., 2020). The progressive refinement strategy is aligned with previous research supporting that when the AI suggests and the user refines, it enables the user to construct new meaning from their ideas (Reza et al., 2025; Wan et al., 2024). This progressive-refinement reflects a trial-and-error process aligned with monitoring and iterative goal adjustment – key features respectively for creative metacognition and self-regulation (Lebuda & Benedek, 2023; Zielińska et al., 2022). It also closely resembles what was described as experimentation strategies, in which multiple approaches, formulas, or tactics are tested to identify what works and how (Jankowska et al., 2018).

Selective appropriation involves reflective evaluation and decision-making about which AI-generated elements align with the user's creative intent. This strategy is in line with the results from Newman and colleagues (2024) which found that GenAI can support the development of creative self-efficacy, particularly when users engage in iterative adjustments. This parallels metacognitive evaluation processes such as assessing outcomes, monitoring progress, and aligning outputs with goals (Fayena-Tawil et al., 2011; Jankowska et al., 2018).

Finally, counter-inspiration – the deliberate rejection of AI suggestions in order to assert originality – appears similar to metacognitive strategies like perspective shifting and is conceptually close to discarding memory-based ideas (Gilhooly et al., 2007). It was also found to be a strategy used in co-creative writing (Hansen & Rafner, 2025; Reza et al., 2025) where the user is actively deviating from the LLM outputs. Although one might be tempted to consider it specific to human-AI collaboration, it is important to note that this may stem from the limited empirical understanding of metacognitive and self-regulatory strategies in collaborative human-human creative contexts. While it could indeed represent a response unique to working with AI, it is also plausible that similar behaviors might be observed in group-based creative problem solving. This calls for further investigation.

The indicated three strategies serve to manage the co-creative dynamic, support effective task performance, and sustain a sense of creative agency. Rather than

replacing familiar cognitive strategies, working with generative systems appears to recontextualize them within new interactional and technological frameworks.

Ownership and other ethical dilemmas

Control and ownership emerged as interlinked dimensions of agency in our study. Participants' reflections revealed a fluid and evolving sense of ownership, echoing previous findings from AI-assisted scriptwriting, where users alternated between feeling like sole authors and perceiving the AI as a "ghostwriter" (Weber et al., 2024). Similar tensions have been noted in AI-assisted creative writing, where authors grapple with whether AI contributions should be credited as co-authorship or treated as editorial support (Chakrabarty, Padmakumar, et al., 2024). One participant asked, "How much would I be taking from the tool suggestions for it to be co-authorship?" (S2P4). These reflections illuminate the dynamic negotiations of ownership that participants undertake during human-AI collaboration. However, this view may not generalize to professional creators or align with legal frameworks such as copyright law, where ownership is typically defined more categorically (Do et al., 2025). Our findings should be interpreted as context-specific and exploratory rather than representative of broader normative positions.

New evidence reinforces the idea that ownership is process- and context-dependent. A recent synthesis and interview study with writers (Reza et al., 2025) shows that what writers want to "own" depends on both the writing process and the writing context. Content-centric writers (e.g., academic or expository) prioritize ownership over planning (ideation, goal-setting), whereas form-centric writers (e.g., literary/creative) insist on ownership during translating and revising (style, voice, sentence-level choices). Monitoring – overseeing when and how AI participates – emerges as a distinct locus of agency that is under-supported in current tools; writers explicitly ask for mechanisms to modulate AI involvement over time (e.g., turning help on/off, adjusting initiative), to preserve both control and ownership across phases.

This also echoes findings in AI-generated art, where artists were divided between attributing ownership to themselves or acknowledging AI's role, particularly when the output closely resembled their original work (Lovato et al., 2024). These findings highlight that when AI outputs align with users' stylistic choices or intentions, their sense of ownership increases (COFI framework, task contribution similarity). Our results regarding the abstractness of images align with broader debates in generative AI art, where realistic outputs

often feel detached from artistic identity, leading to a reduced sense of ownership (Lovato et al., 2024).

Beyond what to own, how systems intervene matters. During writing, interaction features can preserve ownership across processes (Reza et al., 2025). Examples of these interaction features include: suggestion-based help rather than automatic insertion, keeping the final say with the writer, and providing toggles to enable, disable, or scope AI assistance (e.g., allow grammar but block “style fixes” when intentionally breaking rules). In some circumstances multiple suggestions are preferable; the end user can compare and selectively appropriate, instead of single authoritative edits – especially in form-centric work where sentence-level agency anchors authorship (Reza et al., 2025).

Our findings suggest that ownership in human-AI co-creation is not static but shaped by iterative interactions and evolving perceptions of agency. Participants’ sense of ownership often depended on whether they perceived AI as a passive tool or an active collaborator, reinforcing prior studies that emphasize the importance of AI’s role transparency in co-creative processes (Weber et al., 2024). In human-AI co-creative writing systems, users can find it challenging to reengage with the creative process once the AI has significantly contributed (i.e. generated a large portion of text, determined the direction of the plot etc.) which may lead to a reduction in their sense of agency and a tendency to passively accept the AI-generated content (Clark et al., 2018). People relax ownership constraints under high time pressure, low task importance, low confidence in a subskill (e.g., grammar), or higher trust in AI’s reliability. Conversely, high-stakes, identity-expressive tasks trigger narrower delegation and stronger authorial claims (Reza et al., 2025). Future research should further explore how co-creative systems can adapt to varying user preferences and ensure meaningful contributions to the creative process.

The increasing reliance on generative tools raises concerns about the potential for *deskilling*, where frequent AI use diminishes users’ active creative engagement or erodes foundational skills. Rafner et al. (2022) caution that overdependence on AI may lead individuals to outsource core tasks, weakening both competence and confidence over time. This is especially critical in creative domains where process, not just product, matters. Emerging evidence from non-creative domains suggests that AI tools may have an equalizing – or in some cases stratifying – effect depending on users’ baseline proficiency. Studies show that individuals with lower initial performance often benefit most from AI assistance, as seen in customer service (Brynjolfsson et al., 2025) and consulting (Dell’acqua et al., 2023). This points to

a potential “catch-up” effect, where AI serves as a compensatory partner for those with less expertise. However, whether this effect extends to open-ended creative work remains an open question.

Alternatively, human-AI collaboration may benefit all users equally (“a rising tide lifts all boats”) or may exacerbate performance gaps if only experts can critically evaluate AI outputs (“the rich get richer”) (DiStefano et al., 2025; Rafner, Beaty, et al., 2023). For example, in a study of materials scientists, Toner-Rodgers (2024) found that while AI support increased overall innovation, high-performing individuals nearly doubled their output, whereas those with lower expertise saw minimal improvement. These findings suggest that domain knowledge is critical for leveraging AI effectively, raising important questions about fairness and long-term skill development in co-creative AI systems. Because ownership preferences and acceptable delegation vary across process and context, tools that let users selectively delegate (and audit those delegations) may mitigate deskilling by keeping decision-making, monitoring, and core authorial moves with the human (Reza et al., 2025).

Limitations and future research directions

This preliminary study’s small, diverse sample yielded rich insights but limits generalizability. As this is a preliminary study, we do not claim saturation or population-level transferability. Participants differed in creative experience and AI familiarity, which likely shaped their agency strategies. We acknowledge that discount usability does not substitute for saturation in qualitative research. Future work should involve larger samples to deepen theoretical robustness and explore how agency varies by tool, domain, and user background.

Differences in participants’ AI familiarity and creative experience, especially between experienced writers and those with broader creative backgrounds, suggest that future work should investigate how individual traits (e.g., expertise, creative confidence) shape agency in human-AI co-creation. Due to the considerable variation in participants’ prior experience with both AI and creative work, we were unable to use these characteristics as consistent analytical categories. No clear patterns emerged within our sample that would allow for meaningful comparisons along these dimensions. However, we do not rule out the possibility that future studies with larger, purposefully selected groups – differentiated, for example, by level of AI familiarity, type of creative practice, or degree of

expertise in a given domain – could reveal meaningful differences in how agency is expressed, as well as domain-specific patterns of creative engagement (McCarthy et al., 2006).

While this study captures participants' initial interactions with the AI system, the observation that users developed strategies over time suggests the presence of learning effects. Previous longitudinal studies with teachers have shown that individuals tend to develop greater self-regulation and agency over time (Anderson, 2025). Therefore, we believe that future longitudinal research on human – AI collaboration is needed to examine how sustained use of such systems affects users' sense of agency; for example, whether increased familiarity with AI enhances AI-related agency and, in turn, fosters greater creative agency (Watkins & Barak-Medina, 2024).

Findings are also tool-specific, reflecting the distinct affordances of slider-based image generation, text-to-image generation, and Sudowrite. Broader studies across varied AI tools and creative domains are needed to assess how design choices affect autonomy, control, and other dimensions of agency. At theoretical level it would be worthwhile to further develop distinctions between the four agency constructs: For example by increased focus at macro-level creative direction and micro-level control mechanisms in participants' interactions with AI (cf. Botella et al., 2018). The macrolevel refers to overarching decisions – such as setting creative goals, determining the tone or theme of a project, or choosing the target audience. In contrast, the micro-level involves operational strategies, including selecting between AI-generated alternatives, rephrasing outputs, modifying prompts, or blending different suggestions from the system. For example, a participant might decide that a project should convey an ironic message about social media (macro-level direction), while also revising specific sentences generated by the AI to better align with that tone (micro-level control). Future studies could explore how individuals dynamically shift between these levels of agency when collaborating with generative AI systems.

Conclusion

As human-AI co-creative practices gain traction across creative domains, this study provides preliminary insights into how agency is experienced and negotiated when individuals collaborate with AI tools for image generation and creative writing. We identify four dimensions – creative self-confidence, control over creative action, autonomy in the creative process, and

ownership of the creative product – that shape participants' sense of agency throughout the creative process. These findings illuminate the dynamic and context-dependent nature of agency when co-creating with AI tools.

Our analysis highlights how participants adopted and adapted strategies to optimize their task performance, when being challenged by limited control over tool functions or unexpected outputs. These strategies are summarized as *selective appropriation*, *progressive refinement*, and *counter inspiration*. By integrating theoretical insights with practical observations, we demonstrate how agentic creative self-beliefs are expressed and fluctuate throughout the creative process (RQ1), and how participants regulate these beliefs through self-regulation and metacognitive strategies (RQ2) in relation to tool-specific affordances and user-driven approaches. These preliminary findings contribute to a deeper understanding of agency in human-AI co-creation, emphasizing its dynamic and fluctuating character.

Notes

1. Text-to-image generation requires users to write a prompt, describing what they want to depict on the image. It then takes some time to display the result(s).
2. <https://rambollxact.com>
3. <https://www.taguette.org/>
4. Jasper AI is a commercial AI writing platform that utilizes natural language processing to assist users in generating customized written content.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve readability and language, remove redundancies, and shorten text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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